Journal of Memory and Language 150 (2026) 104770



Contents lists available at ScienceDirect

Journal of Memory and Language

journal homepage: www.elsevier.com/locate/jml





The effects of exposure variability and similarity on talker-independent adaptation to Spanish-accented English

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ARTICLE INFO

Dataset link: https://osf.io/z3q6v

Keywords:
Adaptation
Generalization
Perceptual learning
Foreign-accented speech
Word recognition

ABSTRACT

The influence of a talker's first language (L1) on the production of their second (L2) can complicate the task of word recognition. Nevertheless, previous research has shown that listeners demonstrate remarkable speed and plasticity in adapting to L2-accented talkers. Listeners are also able to generalize this learning to novel L2-accented talkers. There are two competing explanations for these talker-independent adaptation effects: one argues that increasing variation during exposure facilitates generalization to a novel test talker (exposure-to-variability hypothesis), while the other argues that increasing acoustic-phonetic similarity between the exposure and test talkers facilitates generalization (similarity-based hypothesis). We conducted three experiments to compare the effects of exposure variability and exposure-test similarity on generalization. During exposure, monolingual English participants performed an auditory lexical decision task with multisyllabic English tokens spoken by three Spanish-accented talkers. After exposure, we tested performance on a novel Spanish- accented talker with a cross-modal matching task (Experiments 1 and 2) or a primed cross-modal lexical decision task (Experiment 3). Exposure variability increased both identity and competitor priming, consistent with a relaxation of phonetic categorization criteria. Exposure-test similarity decreased competitor priming without changing identity priming, while exposure uniformity increased identity priming without changing competitor priming, providing some evidence of phonetic recalibration in these conditions. Our results provide equivocal support for the similarity-based hypothesis and against the exposure-to-variability hypothesis.

Introduction

The production of a single speech sound varies both diachronically within an individual talker and synchronically across groups of talkers. Listeners must accommodate both types of variation in the speech signal in order to achieve successful comprehension. Perceptual adaptation is the process of detecting the *patterns* of variation in the speech signal and applying them during comprehension. This learning process allows listeners to map unfamiliar, variable, or secondary acoustic cues onto phonetic category representations. Previous research has shown that perceptual adaptation to specific cue-category mappings can transfer from one talker to another (e.g., Kraljic & Samuel, 2006, 2007; Reinisch & Holt, 2014). In addition, the benefits of exposure to an L2 accent can transfer to a new talker with the same accent. However, there is disagreement in the literature about the conditions under which adaptation generalizes across talkers. Two different factors have been argued to facilitate generalization: (1) *variability* during exposure (Baese-Berk et al., 2013; Bradlow & Bent, 2008) and (2) *similarity* between the exposure and test talkers (Xie et al., 2021; Xie & Myers, 2017). This paper reports three experiments that directly compared variability and similarity during exposure to Spanish-accented English. The overarching goal is to understand how listeners use acoustic-phonetic information to accommodate linguistic variation.

L2-accented speech

During L2 acquisition, a speaker's L1 phonetic inventory shapes the production of L2 speech sounds (Flege & Bohn, 2021; Nagle & Baese-Berk, 2022). We focused on three contrasting stop consonant pairs found in both English and Spanish: /p/-/b/ (*park* vs. *bark*), /t/-/d/ (*tune* vs. *dune*), and /k/-/g/ (*coal* vs. *goal*). Within each pair, one member differs from the other in terms of voicing (voiceless vs. voiced). Voicing

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https://doi.org/10.1016/j.jml.2026.104770
Received 1 April 2025; Received in revised form 14 May 2026; Accepted 19 May 2026
Available online 30 May 2026
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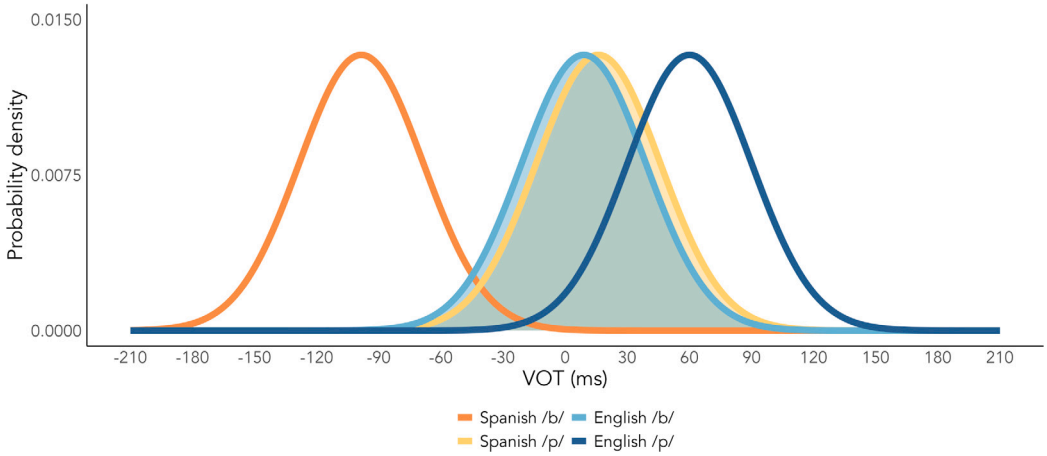


Fig. 1. Overlapping VOT distributions for Spanish voiceless stops and English voiced stops.

refers to the status of the glottis, which comprises two folds of tissue located in the larynx at the top of the neck. Constricting the vocal folds causes them to vibrate, resulting in voiced speech sounds; by contrast, opening the vocal folds limits vibration, resulting in voiceless speech sounds. The primary cue to word-initial stop consonant voicing in both English and Spanish is voice onset time (VOT), which is the interval between the release of the closure of the articulators, called the burst, and the beginning of voicing (Cho & Ladefoged, 1999; Chodroff et al., 2019; Lisker & Abramson, 1964). Listeners are highly sensitive to the probabilistic distributions of VOT across voiced and voiceless stops in their language (Clayards et al., 2008).

The associations between VOT and stop consonants vary cross-linguistically (Chodroff et al., 2019; Lisker & Abramson, 1970). In Spanish, lead (negative) VOTs correspond to voiced stops and short lag VOTs correspond to voiceless stops (Vicente, 1986; Williams, 1977). For example, the /b/ in Spanish *barco* (boat) would be approximately −98 ms on average, while the /p/ in Spanish *parque* (park) would be approximately 16 ms on average (values calculated from Chodroff et al., 2019). By contrast, in English, short lag VOTs correspond to voiced stops and long lag VOTs correspond to voiceless stops (Chodroff & Wilson, 2017). For example, the /b/ in English *bark* would be approximately 9 ms on average, while the /p/ in English *park* would be approximately 60 ms on average (values calculated from Chodroff et al., 2019). As a result, the VOT for Spanish /p/ is a better fit for the contrasting English /b/ phoneme category than for the corresponding English /p/ phoneme category. The overlap in VOT between Spanish /p/ and English /b/ is illustrated in Fig. 1, with probability densities constructed from the data presented in Chodroff et al. (2019).

To illustrate how cross-language transfer in production introduces ambiguity in perception, consider an L1 Spanish speaker who is learning English as an L2. She already has an established /p/-/b/ contrast in her L1 Spanish that is associated with VOT. The /p/-/b/ contrast in her L2 English is associated with the same cue in the same direction, such that VOTs for /p/ are longer than those for /b/. As a result, the English /p/-/b/ contrast will be relatively easy to integrate into her existing phonological system compared to, for example, certain vowel contrasts in English (Baigorri et al., 2019). The same will be true for both the /t/-/d/ and /k/-/g/ contrasts. However, assimilating the L2 English /p/ to the L1 Spanish /p/ means that the English phoneme will be produced like the Spanish phoneme. That is, L2 English /p/ will be produced with a short lag VOT (similar to Spanish /p/). This will create a perceptual problem from an L1 English listener's perspective, since short lag VOTs are associated with English /b/, not /p/; without supporting contextual information, listeners may confuse Spanish-accented English /p/ with /b/ (e.g., *park* perceived as *bark*). To summarize, cross-language transfer in Spanish-accented English can create ambiguity between voiced and voiceless stops (Flege & Eefting, 1987). Listeners are adept at resolving such ambiguities through perceptual adaptation.

Adaptation to L2-accented speech

Listeners rapidly fine-tune their phonetic categories in response to L2-accented speech (Bent & Baese-Berk, 2021). A typical experimental design features an exposure phase, in which participants gain experience with a speech pattern, followed by a test phase. Talker-specific adaptation is the result of hearing the same talker during exposure and test (Clarke & Garrett, 2004; Xie et al., 2017). Talker-independent adaptation, or generalization, is the result of exposure to a *type* of talker rather than exposure to a *specific* talker (Alexander & Nygaard, 2019; Bradlow et al., 2023; Bradlow & Bent, 2008; Sidaras et al., 2009; Tzeng et al., 2016; Xie et al., 2021, 2018). Overall, experience with L2-accented talkers improves perception of novel L2-accented talkers (Baese-Berk et al., 2013; Bieber & Gordon-Salant, 2022; Reinisch & Holt, 2014; Tzeng et al., 2024; Witteman et al., 2013). The pattern of results for talker-independent adaptation becomes more complicated when considering different types of exposure to an accent. Studies often compare single-talker exposure, in which one talker produces all of the stimuli for a condition, to multi-talker exposure, in which several different talkers produce the stimuli for a condition. The literature is divided as to whether multi-talker exposure is necessary for generalization (Baese-Berk et al., 2013; Bradlow et al., 2023; Bradlow & Bent, 2008; Xie et al., 2021; Xie & Myers, 2017).

Two experiments highlight the necessity of multi-talker exposure for talker-independent adaptation. In Bradlow and Bent (2008), Experiment 2), participants completed a sentence transcription task during both exposure and test, and performance was measured in terms of sentence recognition accuracy. There were four exposure conditions: talker-specific, multi-talker, single-talker, and control. Participants in the talker-specific condition were trained directly on the Mandarin-accented English test talker, while those in the multi-talker condition were trained on five novel Mandarin-accented English talkers. Participants in the single-talker condition were exposed to just one of the Mandarin-accented English talkers from the multi-talker condition. Control training featured five L1-accented English talkers. Training was followed by two post-tests: one featured a novel talker with a familiar L2 accent (Mandarin) and the other featured a novel talker with an unfamiliar L2 accent (Slovakian). Performance on the Mandarin-accented English test talker was higher in the talker-specific and multi-talker conditions than in the single-talker and control conditions, which did not differ from one another. By contrast, performance on the Slovakian-accented English test talker did not differ between conditions. These results suggest that exposure to multiple talkers with the same L1 highlights the features that characterize a specific L2 accent.

Baese-Berk et al. (2013) provided evidence that exposure to multiple L2-accented talkers with *different* L1s also facilitates generalization. They used the same stimulus materials and procedures as Experiment

2 in Bradlow and Bent (2008) in order to compare a multi-accent exposure group directly to the multi-talker exposure and control groups from the previous study. During multi-accent training, participants were exposed to five L2-accented English talkers, each of whom had a different L1: Thai, Korean, Hindi, Romanian, and Mandarin. Multi-accent and multi-talker exposure yielded higher performance on the Mandarin-accented test talker than control exposure. Multi-accent exposure also generalized to the Slovakian-accented test talker, while multi-talker and control exposure did not. Taken together with the results of Bradlow and Bent (2008), these findings suggest that exposure to multiple L2-accented talkers facilitates talker-independent adaptation to a greater extent than exposure to a single L2-accented talker.

By contrast, another set of studies found that single-talker exposure is sufficient for generalization (Bradlow et al., 2023; Xie et al., 2021). Xie et al. (2021) specifically sought to replicate (Bradlow & Bent, 2008) with one key difference: counterbalancing the assignment of talkers to the exposure and test phases across participants. This change allowed them to address a potential confound between the type of exposure (multi- vs. single-talker) and the specific talkers used for exposure and test. They replicated the original finding that multi-talker exposure to Mandarin-accented English facilitates generalization more than control exposure. However, contrary to Bradlow and Bent (2008), they found that single-talker exposure also facilitated generalization more than control exposure. When comparing multi- and single-talker exposure, the difference in performance depended on the particular combination of exposure and test talkers. Bradlow et al. (2023) also observed asymmetries in performance between single- and multi-talker exposure when rotating four talkers, each with different L1s, through the exposure and test roles. Interestingly, they observed these asymmetries even when the test talker was an exposure talker. Overall, Bradlow et al. (2023) concluded that single-talker exposure can be sufficient for generalization when the talkers have the right combination of exposure-test similarity and baseline intelligibility.

Xie and Myers (2017) found that generalization is strongly influenced by talker-specific features. Since the design of the present study was based on that of Xie and Myers (2017), we will describe the paradigm in detail here. During exposure, participants performed an auditory lexical decision task with multisyllabic items produced by five Mandarin-accented English talkers (Experiment 1). In two follow-up studies, participants performed the same exposure task with only one Mandarin-accented English talker (Experiments 2 and 3). Critically, participants were randomly assigned to one of two exposure groups that featured different sets of real words on critical trials. Experimental exposure included /d/-final words (e.g., *overload*), while control exposure included words without /d/ in any position (e.g., *animal*). The exposure manipulation focused on word-final /d/ because it is perceptually confusable with /t/ in Mandarin-accented English (e.g., *seed* perceived as *seat*). During test, participants performed a primed cross-modal lexical decision task, in which auditory primes produced by a novel Mandarin-accented talker were followed by visual targets. On critical trials, the visual targets were either /d/-final real words or their /t/-final minimal pairs (e.g., seed or seat). The priming effect, or the difference in response times between trials with matching /d/-final auditory primes (e.g., seed) and those with unrelated auditory primes (e.g., fair), indexed the amount of talker-independent adaptation.

The multi-talker exposure results indicated that listeners shifted their phonetic boundaries with exposure to word-final /d/ in line with other work on generalization. Specifically, the experimental group exhibited larger priming effects for matching /d/-final targets than the control group. This implies that auditory *seed* became a better prime for visual *seed* after exposure to disambiguating lexical contexts, suggesting that exposure increased activation of the intended word. The experimental group also exhibited larger priming effects for matching /d/-final targets than for competing /t/-final targets, while the magnitude of these priming effects did not differ in the control group. This implies that auditory *seed* became a worse prime for visual *seat* after exposure to disambiguating lexical contexts, suggesting that exposure decreased lexical activation of the unintended word. In other words, exposure reduced competition between /d/- and /t/-final minimal pairs. If listeners in the experimental group had simply expanded their phonetic category for /d/, the priming effects for /d/-final targets would still have been larger than those in the control group (indicating increased activation of the matching targets), but the priming effects for /t/-final targets would have also been larger (indicating increased activation of the competing minimal pairs). Overall, the results showed that exposure to word-final /d/ in disambiguating lexical contexts like *overload* reshaped listeners' phonetic categories for /d/ and /t/, such that the auditory prime *seed* was not only perceived more like the matching word *seed*, but also perceived less like its minimal pair *seat*.

The single-talker exposure results demonstrated that multi-talker exposure is not necessary for generalization. Two of the five exposure talkers from the main experiment were selected for single-talker exposure: one who was the most similar to the test talker on three key acoustic measures of word-final /d/ and one who was the least similar. When the exposure talker was similar to the test talker, the experimental group exhibited larger priming effects for /d/-final targets than for /t/-final targets, while the control group did not. By contrast, when the exposure talker was dissimilar to the test talker, there were no differences between exposure groups or target types. These findings suggest that exposure to a single talker with similar word-final /d/ realizations to the test talker reduced competition between /d/- and /t/-final minimal pairs. Moreover, this reduction in lexical competition with a single similar exposure talker was comparable to the one observed in the main experiment with multiple exposure talkers. Together, the results of Xie and Myers (2017) show that the correspondence between exposure and test talkers at the acoustic-phonetic level is critical for understanding the observed patterns of generalization across studies.

Competing hypotheses for generalization

There are two competing explanations for why multi-talker exposure may or may not provide additional benefits for generalization over single-talker exposure. The **exposure-to-variability hypothesis** posits that L2-accented talkers exhibit similarities in production that differ from L1-accented norms (Baese-Berk et al., 2013). The exposure-to-variability hypothesis also posits that among L2-accented talkers with the *same L1*, cross-language influence shifts the relations between acoustic cues and phonetic categories similarly across talkers. Among L2-accented talkers with *different L1s*, typological features that are unique to the L2 lead to similarly accented realizations regardless of the L1. Multi-talker exposure thus allows listeners to abstract away from the peculiarities of any given talker and home in on these commonalities. For example, a listener encountering one unfamiliar Spanish-accented English talker may not know whether their short lag VOTs are specific to that talker or characteristic of the Spanish accent. By contrast, a listener encountering multiple unfamiliar Spanish-accented English talkers at once would not only hear that short lag VOTs are common across the talkers, but also that other accent features (e.g., vowel height) exhibit covariation with voicing (Clayards, 2017). In Bradlow and Bent (2008), multi-talker exposure outperformed single-talker exposure, suggesting that training on multiple L2-accented talkers (with the same L1) allowed listeners to separate the characteristic features of an accent from the idiosyncratic features of a talker. The results of Bradlow and Bent (2008) support the exposure-to-variability hypothesis, but do not align straightforwardly with the effects observed by Xie et al. (2021), Xie and Myers (2017), or Bradlow et al. (2023).

The **similarity-based hypothesis** posits that acoustic-phonetic overlap between the exposure and test talkers, rather than variability during exposure, facilitates generalization (Xie et al., 2021). In Xie and

Myers (2017), listeners exhibited comparable talker-independent adaptation effects after both single- and multi-talker exposure to Mandarin-accented realizations of word-final /d/, which is perceptually confusable with /t/ (e.g., *seed* vs. *seat*). Critically, both exposure conditions contained a Mandarin-accented talker with similar word-final /d/ acoustics to the test talker. Xie et al. (2021) also demonstrated equivalent generalization effects from single- and multi-talker exposure to Mandarin-accented English. They argue that exposure to multiple talkers merely increases the likelihood that listeners will encounter a cue distribution that is relevant for adapting to the test talker. For example, multi-talker training on Spanish-accented speech is more likely to include at least one talker with short lag VOTs than single-talker training. Bradlow and Bent (2008)'s multi-talker exposure condition may have included talkers who were more similar to the test talker than their single-talker exposure condition, which would confound talker similarity and exposure to variability. Bradlow et al. (2023) and Xie et al. (2021) addressed this confound by counterbalancing the combinations of exposure and test talkers across participants. Thus, the similarity-based hypothesis may account for the differential effects of single- and multi-talker exposure in perceptual adaptation studies. The present study was designed to distinguish between the roles of variability and similarity in talker-independent adaptation. To summarize, the similarity-based hypothesis focuses on specific cue-category mappings, while the exposure-to-variability hypothesis focuses on the covariation between cues.

Comparing variability and similarity

As discussed above, the results of Bradlow and Bent (2008) and Baese-Berk et al. (2013) support the exposure-to-variability hypothesis, while those of Xie and Myers (2017) and Xie et al. (2021) support the similarity-based hypothesis. The results of Bradlow et al. (2023) fell in between, with adaptation depending greatly on the specific combinations of talkers. We argue that the ideal adapter framework provides a link between these two hypotheses and can account for these conflicting findings (Kleinschmidt, 2019; Kleinschmidt & Jaeger, 2015). Under this theory, listeners are sensitive to the probability distributions of cue-category mappings—like the ones shown in Fig. 1 for the acoustic cue VOT over the phonetic categories /p/ and /b/—and either create new or updated existing mental representations of these distributions through statistical learning. The specificity of these cue-category representations, called generative models, depends on two factors: how much information is gained about a cue distribution and how well a category predicts a cue by using a particular grouping. An ideal adapter will structure her generative models according to the most "informative" and "useful" socio-indexical groupings (Kleinschmidt, 2019). For L2-accented speech, these groupings may be structured according to each talker (talker-specific) or to the L2 accent shared by the talkers (talker-independent). The socio-indexical structure of cue-category mappings is critical for reconciling the exposure-to-variability and similarity-based hypotheses.

The exposure-to-variability hypothesis argues that increasing variability during exposure increases systematic covariation among relevant cues, enabling listeners to separate the idiosyncratic features of a talker from the common features of an accent. Similar arguments have been made regarding high variability phonetic training paradigms for non-native contrasts (Brekelmans et al., 2022). To restate this hypothesis in terms of the ideal adapter framework, covariation among individual exposure talkers promotes the development of a robust talker-independent model (or refinement of an existing one). In turn, this talker-independent model guides adaptation to the novel test talker. This exposure-to-variability perspective assumes that talker-independent models are better for generalization than talker-specific models.

By contrast, the similarity-based hypothesis argues that increasing the similarity between the cue-category mappings of the exposure and test talkers facilitates generalization. Translating this hypothesis into the ideal adapter framework, listeners develop robust talker-specific models during exposure. During test, listeners use the model that provides the most information about the test talker's cue distributions and most readily predicts their individual cue values to guide adaptation. This similarity-based perspective assumes that talker-specific models are best for generalization because they provide precise information about cue-category mappings.

Overall, the specificity of a listener's set of generative models may explain the effects of variability and similarity on generalization. The present study tests these two hypotheses in order to probe the mechanisms underlying adaptation to L2-accented speech. The overarching goal is to understand how the L1 speech recognition system learns the patterns of cross-language influence in L2 speech production.

Present study

How do listeners generalize their experience with L2-accented speech? On the one hand, the *structure* of exposure may be the primary driver of talker-independent adaptation. Variability has been the primary focus of structural inquiries (Baese-Berk et al., 2013), with number of talkers being the key manipulation (Bieber & Gordon-Salant, 2022; Bradlow & Bent, 2008; Choi & Perrachione, 2019; Xie et al., 2021). On the other hand, the *content* of exposure may be the key to effective generalization. At a high level, training on the relevant L2 accent tends to facilitate talker-independent adaptation (Alexander & Nygaard, 2019; Bradlow et al., 2023; Clarke & Garrett, 2004; Xie et al., 2018). At a detailed level, exposure to the relevant acoustic-phonetic features of an L2 accent explains generalization beyond shared L1 (Reinisch & Holt, 2014; Sidaras et al., 2009; Xie & Myers, 2017). Together, a shared L1 and common acoustic-phonetic features create similarity between exposure and test talkers that facilitates adaptation. The present study builds on this body of work by investigating generalization of L2-accented speech. The key difference between our study and earlier work is the operationalization of variability and similarity. Specifically, we increased the precision with which variability was implemented (cf. Baese-Berk et al., 2013) and expanded the scope of similarity (Bradlow et al., 2023; cf. Xie & Myers, 2017) in order to better delineate the exposure-to-variability and similarity-based hypotheses. The goal was to clarify some of the inconsistent findings in the literature for talker-independent adaptation to L2-accented speech (Bent & Baese-Berk, 2021).

Regarding variability, previous studies have almost exclusively investigated this factor as a comparison between single-talker and multi-talker exposure (cf. Baese-Berk et al., 2013; Bradlow et al., 2023). However, multi-talker exposure reduces the amount of experience with any given talker relative to single-talker exposure. From the larger perceptual adaptation literature, we know that listeners are highly sensitive to acoustic-phonetic mappings and use them to adapt to new talkers (e.g., Kraljic & Samuel, 2006). We also know that there is a high degree of within- and between-talker variability in both L1- and L2-accented speech production (Wade et al., 2007; Xie & Jaeger, 2020). Depending on the idiosyncrasies of a given exposure talker, altering the amount of experience with this talker may benefit or impede generalization. These asymmetries in talker-specific exposure may explain some of the inconsistent findings in the literature (Bradlow et al., 2023; Xie et al., 2021; Xie & Myers, 2017). Moreover, exposure to multiple talkers increases the sources of covariation to account for, which in turn increases the difficulty of accounting for them during exposure. If we assume, as the exposure-to-variability hypothesis does, that experience with covariation is critical for generalization, this lack of clarity obscures how and why variability might facilitate generalization. In the present study, we used a single group of talkers with the same L1 to control the type and amount of experience with each talker across levels of variability.

Regarding similarity, previous studies have taken two general approaches. One focuses on the L1 of the exposure and test talkers, with similar talkers sharing an L1 and dissimilar talkers having different L1s (e.g., Bradlow & Bent, 2008). Another focuses on the acoustic-phonetic features of the exposure and test talkers, with similar talkers having greater overlap in their distributions than dissimilar talkers (e.g., Xie & Myers, 2017). To operationalize similarity, both approaches use different talkers for each exposure condition (rendering the same problem we discussed above for multi-talker vs. single-talker exposure). As a result, we cannot know whether listeners are "learning a talker or learning an accent" or learning both Xie and Myers (2017). In other words, this design confounds low-level talker-specific effects and high-level accent-specific effects. Here, we manipulated the stimuli listeners encounter during exposure rather than the talkers in order to maintain consistency across levels of similarity. Moreover, we directly crossed the factors of similarity and variability in the same design, which, to our knowledge, has not yet been done. Overall, these manipulations provide a novel way to address the ongoing debate between the exposure-to-variability and similarity-based hypotheses for generalization. A summary of the experimental design in relation to our research questions is presented in the next section. The data and analyses for all three experiments can be found on the Open Science Framework (https://osf.io/z3q6v).

General methods

We conducted three experiments with an exposure-test design to understand how different kinds of experience with Spanish-accented speech change listeners' VOT-stop mappings. The exposure phase established the comparison between the similarity-based and exposure-to-variability hypotheses of talker-independent adaptation. Participants performed an auditory lexical decision task with a between-subjects manipulation that exposed them to different sets of multisyllabic real words and pseudowords (Xie & Myers, 2017). In Experiment 1, there were six between-subjects conditions. In Experiments 2 and 3, two of these conditions were removed, and changes to the pseudowords were made to improve learning.

The test phase assessed the effects of each type of exposure on perception. In Experiments 1 and 2, participants performed a cross-modal matching task with a within-subjects manipulation that tested a common set of monosyllabic real words. In Experiment 3, participants performed a cross-modal primed lexical decision task that better assessed learning from exposure. Both tasks featured auditory primes and visual targets and were adapted from Xie and Myers (2017). Across experiments and phases, participants heard four Spanish-accented talkers—three during exposure and one during test—with the talker's role counterbalanced across participants. We also conducted a norming study to select the items for both tasks, which can be found in Appendix A.

Stimulus design

To clearly describe the overall experimental design, it is necessary to first describe the design of the stimuli. Stimuli were grouped by onset phoneme (e.g., *park* has the onset phoneme /p/). There were three **experimental onset** groups: critical, competitor, and control. Critical onsets were voiceless stops (/p/, /t/, and /k/), competitor onsets were voiced stops (/b/, /d/, and /g/), and control onsets were voiceless fricatives (/f/, /s/, and /ʃ/). Across the experimental onset groups, phonemes were also grouped (roughly) according to their place of articulation: labial (/p/, /b/, and /f/), alveolar (/t/, /d/, and /s/), and postalveolar/velar (/k/, /g/, and /ʃ/). These will be referred to as the **cross-experimental onset** groups. There was one **filler onset** group: /m/, /n/, /l/, /ɹ/, /h/, and /w/. Filler onsets were also grouped by their place of articulation: nasal (/m/ or /n/), alveolar (/l/ or /ɹ/), and other back consonants (/h/ or /w/). These will be referred to as the **within-filler onset** groups. Finally, each cross-experimental onset group was paired with one of the within-filler onset groups according to frontness/backness: front (/p/, /b/, /f/, /m/, and /n/), mid (/t/, /d/, /s/, /l/, and /ɹ/), and back (/k/, /g/, /ʃ/, /h/, and /w/). These will be referred to as the **cross-condition onset** groups.

Exposure phase design

Exposure similarity was operationalized as the relation between the experimental onsets encountered during exposure and the critical onsets encountered during test. In Experiment 1, there were three levels of Similarity: Direct, Indirect, and Control. Experiments 2 and 3 included the Direct and Indirect levels only. Each level refers to the type of information participants received about Spanish-accented voiceless stops. In Direct conditions, participants were exposed directly to critical onsets (e.g., *peanut*). In Indirect conditions, participants were exposed to competitor onsets (e.g., *beehive*), thereby gaining experience with the shifted VOT continuum that they would encounter during test. In Control conditions, participants were exposed to control onsets (e.g., *football*), thereby gaining general experience with Spanish-accented English but not with stop VOTs. This design allowed the talkers to remain the same across levels of Similarity. The top of Fig. 2 provides examples of the real words that comprised each level of Similarity; the examples were selected in order to illustrate the three cross-experimental onset groups that were included across levels (front, mid, and back/other).

Exposure variability was operationalized as the relation between onset phonemes and exposure talkers. There were two levels of Variability: Invariant and Variant. Each level refers to the type of experience with each talker. In Invariant conditions, listeners heard each of the three exposure talkers produce one onset out of the three in an experimental group (e.g., Direct-Invariant: Talker A produced *peanut*, *palace*, and *pianist*; Talker B produced *terminal*, *tiara*, and *textile*; and Talker C produced *kingdom*, *counter*, and *kayak*). This means that all of the words with a given onset were produced by only one talker. In Variant conditions, listeners heard each of the three exposure talkers produce all three onsets in an experimental group (e.g., Direct-Variant: Talker A produced *peanut*, *terminal*, and *kingdom*; Talker B produced *palace*, *tiara*, and *counter*; and Talker C produced *pianist*, *textile*, and *kayak*). This means that one third of the words with a given onset were produced by each talker. This design allowed the items and talkers to remain the same between levels of Variability. The two levels of Variability are depicted at the top of Fig. 2, with green arrows/boxes illustrating Variant talker assignments and purple arrows/boxes illustrating Invariant talker assignments (the reader is referred to the web version of this article for the interpretation of color references).

Test phase design

There were three levels of Target: Identity, Competitor, and Unrelated. Each level refers to the type of visual target that followed each auditory prime. Identity targets exactly matched the auditory primes (e.g., *park-park*). Competitor targets were the minimal pairs of the auditory primes (e.g., *park-bark*). Unrelated targets only shared vowels with the auditory primes (e.g., *park-wand*). Differential performance on these three conditions indexed learning from exposure. The test phase is depicted at the bottom of Fig. 2 with the auditory prime *park* and its corresponding visual targets. In Experiments 1 and 2, participants made an explicit decision about whether the auditory prime matched the visual target. In Experiment 3, participants made a decision about whether the visual target was a real word or not, with the preceding auditory stimulus acting as a prime for the lexical decision.

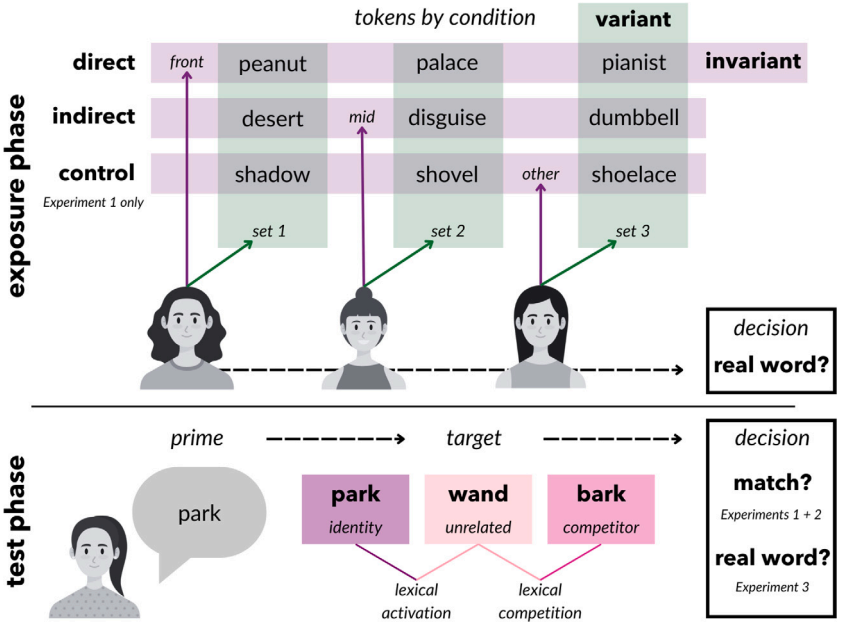


Fig. 2. Cross-experiment design.

Predictions

The exposure-to-variability and similarity-based hypotheses make different predictions about the effects of each type of exposure on test. The test task in Experiments 1 and 2 measures explicit judgments about the match/mismatch between auditory primes and visual targets, while the test task in Experiment 3 measures more implicit priming effects. In both cases, performance on Identity and Competitor targets relative to Unrelated targets indexes changes in perception as a result of exposure. Smaller differences in accuracy and RTs between Competitor and Unrelated targets (*park-bark* vs. *park-wand*) would be evidence of decreased lexical competition. Larger differences in accuracy and RTs between Identity and Unrelated targets (*park-park* vs. *park-wand*) would be evidence of increased lexical activation. Exposure conditions that facilitate perceptual adaptation should elicit both of these patterns, which together indicate that listeners have fine-tuned their phonetic categories to accommodate Spanish-accented stops. The presence of only one of these patterns would provide some evidence of perceptual adaptation. A reversal of the matching/priming effects (i.e., larger differences in performance between Competitor and Unrelated targets and smaller differences between Identity and Unrelated targets) would indicate a general relaxation of phonetic category boundaries rather than perceptual recalibration (Zheng & Samuel, 2020).

The similarity-based hypothesis predicts a main effect of Similarity, with evidence for perceptual adaptation increasing from Control to Indirect to Direct exposure. This prediction follows directly from the ideal adapter framework, such that both talker-specific and talker-independent generative models of VOT-stop distributions can generalize to a test talker; what matters is how similar the exposure model is to the test distribution. As the name implies, Direct exposure provides the most direct experience the VOT distributions that characterize voiceless stops in Spanish; however, Indirect exposure also provides experience with these distributions that listeners may use to generalize to the novel talker. If listeners are sufficiently sensitive to the covariation between voiced and voiceless stop VOTs within talkers, the similarity-based hypothesis would also predict comparable patterns of generalization from both Direct and Indirect exposure (e.g., Kraljic & Samuel, 2006).

The exposure-to-variability hypothesis predicts a main effect of Variability, with more robust perceptual adaptation after Variant exposure than after Invariant exposure. Under this hypothesis, exposure to covariation between cues and categories encourages the formation of talker-independent generative models. This implies that such models have more utility for categorizing critical onsets than any talker-specific models (Kleinschmidt, 2019). The exposure-to-variability hypothesis does not predict effects of Similarity on test performance.

Talkers

Four female Spanish-accented English talkers from Mexico recorded the experimental stimuli (Talkers 1–4). In addition, a female L1-accented English talker from the US recorded the practice stimuli (Talker 0). All four L2-accented talkers reported that they grew up speaking Spanish with their caregivers and began acquiring English in traditional classroom settings. Additional language background information and recording details can be found in Appendix B.

We will briefly describe the VOT distributions of each talker here and refer the reader to Appendix B for details. Talker 4 produced the most Spanish-like VOTs across places of articulation, with shorter means than the other talkers. Talker 3 produced overall longer VOTs than the other talkers, which even fell outside L1 US English norms. Talkers 1 and 2 were the most similar to one another, with Talker 1 aligning the most with L1 English norms. Overall, each talker exhibited a unique pattern of VOT-stop distributions.

Experiment 1: Investigating differences in generalization from exposure to Spanish-accented stops versus fricatives

Methods

Participants

We recruited 296 participants through the online platform Prolific. Participants provided implied consent in line with Penn State IRB policies and were compensated $6 after completing the experiment, which took approximately 30 min. The experiment was available to individuals whose Prolific user profiles aligned with the following eligibility criteria: between 18 and 40 years of age, located in the US at the time of the study, English as their first and only fluent language, normal hearing, normal or corrected-to-normal vision, and without a history of language-related disorders. The Prolific user profiles for each participant also included sex (two options, select one) and race/ethnicity information (four options, select all that apply).

The eligibility criteria were cross-checked with the responses to a post-experiment questionnaire (see Section "Tasks"), and any ineligible participants were removed from further analysis (N = 9). We also removed participants who reported that their L1 or L2 was Spanish, with self-rated proficiency greater than or equal to 3/5 in any languages other than English, with self-rated proficiency less than 5/5 in English, or whose place of origin was not the US (N = 15). This left 287 eligible participants.

Finally, we removed participants with poor data quality from further analysis (N = 15). Poor data quality was defined as: accuracy statistically indistinguishable from or significantly below chance on experimental real words in the exposure task, zero correct experimental trials in any of the three conditions in the test task, or fewer than 50% of experimental trials with correct responses and reaction times between 50 and 2500 ms in either task (see Section "Tasks" for details). This left 257 participants for analysis (Age: M = 31, SD = 6, Min = 18, Max = 40; Sex: Female = 121, Male = 135, Prefer not to say = 1; Race: Asian = 5, Black = 43, Multiple selected = 21, Other = 5, White = 182, Not provided = 1). Our *a priori* power analysis for the 2 x 3 factorial design (Variability x Similarity) indicated that we needed 27 participants per condition, which we achieved ($1-\beta = 0.8$, $\alpha = .05$, and Cohen's $f = 0.25$).

We recruited 46 additional participants through Prolific to complete the experiment without the exposure phase. Removing ineligible participants left 38 participants. Removing participants with low data quality left 37 participants for analysis (Age: M = 31, SD = 5, Min = 21, Max = 40; Sex: Female = 19, Male = 18; Race: Asian = 2, Multiple selected = 5, Other = 2, White = 28). All aspects of recruitment were the same for this group as for the main group of participants.

Stimuli

In total, there were 936 auditory items and 648 visual items across the exposure and test tasks. The final set of exposure items included 360 real words (24 per onset) and 360 pseudowords (24 per onset). The final set of auditory primes included 216 real words (24 per onset). The final set of visual targets included 648 real words (3 per auditory prime). An additional set of six real words and six pseudowords with filler onsets (one per onset per word type) was selected for practice. Six additional primes with filler onsets (one per onset) were selected for practice. Two additional unrelated targets were also selected for practice. Stimulus selection is described in Appendix C.

Experimental lists

Combining Similarity and Variability created six between-subjects conditions: Direct-Variant, Direct-Invariant, Indirect-Variant, Indirect-Invariant, Control-Variant, and Control-Invariant. In addition, one group of participants did not receive any exposure prior to test; this will be referred to as the Test-only condition. The 288 filler items were the same across conditions and evenly divided by word type—real word and pseudoword—as well as by onset—/m/, /n/, /l/, /ɹ/, /h/, and /w/. The 144 experimental items were evenly divided by onset and word type, with onset differing by level of Similarity: Direct (critical: /p/, /t/, and /k/), Indirect (competitor: /b/, d/, and /g/), and Control (control: /f/, /s/, and /ʃ/). As a result, there were 24 items per onset per word type.

The assignment of talkers to items differed by level of Variability. In Invariant conditions, talkers were assigned by cross-condition onset group: front (/p/, /b/, /f/, /m/, and /n/), mid (/t/, /d/, /s/, /l/, and /ɹ/), and back/other (/k/, /g/, /ʃ/, /h/, and /w/). In other words, one talker was assigned to front onsets, one to mid, and one to back/other. Within a given level of Similarity, this means that each talker produced items with one out of the three experimental onsets and two out of the six filler onsets. For Variant conditions, one third of each of the items of a given onset and word type (8) was randomly assigned to one of three sets: set 1, set 2, or set 3. Each set was assigned to one talker.

To counterbalance which talkers were assigned to which items across participants, each Variant item set was paired with one Invariant onset group to create three assignment groups: group 1 (front with set 1), group 2 (mid with set 2), and group 3 (back/other with set 3). In addition, we also counterbalanced the talkers between the exposure and test phases. To do this, we added group 4 (test) and rotated the four talkers across these four assignment groups in a Latin square design. Overall, this resulted in 24 experimental lists for the exposure phase, one for each combination of exposure condition (6) and talker assignment (4).

Regardless of exposure condition, all participants heard the same 216 auditory primes during test. One third (72) were critical items divided evenly by onset—/p/, /t/, and /k/—and two thirds (144) were filler items divided evenly by onset—/m/, /n/, /l/, /ɹ/, /h/, and /w/. Each onset was divided evenly by target type—Identity, Competitor, and Unrelated. This resulted in eight items per onset per target type. Three experimental lists were created to counterbalance the combinations of auditory prime and visual target across participants. Since the test talker was also counterbalanced across participants, this resulted in 12 experimental lists for the test phase, one for each combination of prime-target pair (3) and talker assignment (4).

Exposure practice was presented in Talker 0's voice for all participants. Test practice was presented according to the participant's talker assignment and Variability condition. Participants in the Test-only condition completed the test practice in Talker 0's voice.

Tasks

The experiment included a headphone check, exposure task, test task, and post-experiment questionnaire. All aspects of the experiment were conducted online using Pavlovia. The headphone check, exposure task, and test task were created in PsychoPy Builder (Peirce et al., 2019). The post-experiment questionnaire was built using Pavlovia's survey platform.

The headphone check followed the anti-phase tone test procedure from Woods et al. (2017). On each trial, participants listened to three pure tones, one of which was out of phase with the other two. Participants indicated which of the three tones was the quietest by selecting the appropriate button on the screen: tone 1, tone 2, or tone 3. There were two trials per response for a total of six trials, which were presented in random order. Participants using well-functioning headphones should have easily perceived the anti-phase tone as the quietest, while those using loudspeakers should not. Participants completed the task at most two times.

The exposure phase featured the auditory lexical decision task from Xie and Myers (2017). On each trial, participants indicated whether an auditory stimulus was a real English word or not by pressing the *d* or *k* key on their keyboard. One of the two response-key relations—real-*d* or real-*k*—was assigned randomly to each participant. Participants completed 12 practice trials followed by 432 main trials presented in random order. Half of the practice trials (6) and half of the main trials (216) required real word responses.

The test phase featured a cross-modal matching task adapted from the primed cross-modal lexical decision task in Xie and Myers (2017). On each trial, participants first heard a real word (auditory prime) and then saw a real word written on the screen (visual target). They indicated whether the visual target matched the auditory prime or not by pressing the *d* or *k* key on their keyboard. The assignment of responses to keys was carried over from the exposure phase, with *match* responses mapped to the same key as *real word* responses in the exposure task. Participants completed six practice trials followed by 216 main trials presented in random order. Half of the practice trials (3) and one third of the main trials (72) required match responses.

The post-experiment questionnaire included two sets of items. The first set of items related to the talker from the test task. The second set of items included questions about the participant's own language background and demographics. This second set of items was used to confirm the participant's eligibility as described in Section "Participants". Further details about the questionnaire items and descriptive statistics can be found in Appendix D.

Procedure

Eligible participants accessed the experiment through Prolific. Once a participant began the experiment, they were randomly assigned to one of the experimental lists. First, they performed the headphone check. If the participant achieved fewer than five correct trials out of six, they completed the task again. Participants who failed the headphone check a second time were not allowed to continue; instead, they were redirected to Prolific and asked to return their submission. After passing the headphone check, participants continued to the exposure task and completed the practice (participants in the Test-only condition continued straight to the test task). If a participant scored 50% or lower on the exposure practice, they were not allowed to continue; instead, they were redirected to Prolific and asked to return their submission. After successfully completing the practice session, the participant performed the exposure task. Next, the participant continued to the test task, where they completed the practice session and the main task. Immediately following the test task, participants continued to the post-experiment questionnaire. Once the questionnaire was complete, the participant was redirected to Prolific and received compensation.

Analysis approach

Data processing and analysis were conducted with R version 4.5.1 (R Core Team, 2022). Filler items were removed before proceeding with the following data processing pipeline and therefore were not included in any of the statistical analyses.

Prior to analyzing the exposure task data, we removed responses with RTs less than 50 ms (N = 11; 0.03%). RT was calculated from the onset of the word. For the exposure task, analyses were conducted on the accuracy data only.

Prior to analyzing the test task data, we removed responses with RTs less than 50 ms (N = 4; 0.02%). RT was calculated from the presentation of the visual target. For the test task, analyses were conducted on both the accuracy and RT data. For the RT analyses, we filtered for correct responses. We used the *robustbase* package to calculate adjusted boxplot statistics (appropriate for such skewed distributions as reaction time) to detect responses with reaction times outside the upper and lower fences (Hubert & Vandervieren, 2008). We detected and removed outliers by Target type (total N = 898; 4.24%). The upper and lower fence calculations included data from Experiment 2 in order to avoid using different outlier criteria for the Test-only group between experiments.

Mixed-effects models were fitted to trial-level data with the *lme4* package (Bates et al., 2015). Generalized linear mixed-effects models with a binomial distribution and logit linking function were used to analyze binary accuracy (1,0). Generalized linear mixed-effects models with an Inverse Gaussian distribution and identity linking function were used to analyze RT (see Lo & Andrews, 2015). Models were fit with the full random effects structure that was relevant to the task (see below). To test the significance of a fixed effect, Type-III analysis-of-deviance tables were calculated and Wald chi-square tests conducted with the *car* package (Fox & Weisberg, 2019).

Exposure task analyses modeled the fixed effects of Variability, Similarity, Word type, and their interactions on accuracy. The two levels of Variability and Word type were sum contrast-coded. The three levels of Similarity were simple-coded with Direct as the reference level. Scaled VOT, trial, and word frequency were included as covariates. Participant, talker, and item were included as random intercepts. Word type was included as an uncorrelated random slope with participant. Only significant effects involving Word type and the exposure conditions will be discussed.

For the test task, we conducted two analyses. The first modeled the fixed effects of Exposure, Target, and their interaction on accuracy and RT. Exposure had seven levels: one for each of the six exposure conditions and one for the Test-only condition. This factor was simple-coded with the Test-only condition as the reference level. The three levels of Target were simple-coded with Unrelated as the reference level. Scaled VOT, scaled trial, and the interaction between scaled prime frequency and scaled target frequency were included as covariates. Participant, talker, and the interaction between auditory prime and visual target were included as random intercepts. Target was included as an uncorrelated random slope for participant. If the first analysis revealed significant differences between the Test-only condition and the exposure conditions related to the Target contrasts, we conducted the second analysis focusing only on the exposure conditions; these modeled the fixed effects of Variability, Similarity, Target, and their interactions. The Variability and Similarity coding schemes were the same as in the exposure task analyses. The Target coding scheme, covariates, and random effects were the same as in first set of test task analyses. Significant effects involving Target and the exposure conditions will be discussed for the first analysis; only significant effects involving the exposure conditions will be discussed for the second analysis to avoid redundancy.

Table 1
Experiment 1 exposure task performance: Mean percent correct (standard error) by Word type for each combination of Similarity and Variability. Contrast presented in log odds rather than probability to reflect unit of analysis.

Similarity	Variability	N	Real words	Pseudowords	Real words vs. Pseudowords
Direct	Invariant	42	80.17 (1.26)	83.94 (1.39)	−0.257
Direct	Variant	38	78.78 (1.26)	85.84 (1.23)	−0.490
Indirect	Invariant	41	77.84 (1.03)	85.10 (1.60)	−0.486
Indirect	Variant	41	79.26 (1.18)	85.64 (1.22)	−0.445
Control	Invariant	49	79.35 (1.09)	86.71 (1.22)	−0.530
Control	Variant	46	75.56 (1.23)	88.67 (1.42)	−0.929

Results

Exposure

There was a main effect of Word type ($\chi^2(1, N = 2) = 19.75$, $p < .001$) that reflected overall higher accuracy on pseudowords than on real words ($\beta = -0.56$, $SE = 0.13$, $z = -4.44$, $p < .001$). Summary statistics by Word type and exposure condition are shown in Table 1.

Test: Comparison to the test-only group

For accuracy, there were main effects of Target ($\chi^2(2, N = 3) = 200.43$, $p < .001$) and Exposure ($\chi^2(6, N = 7) = 15.81$, $p = .015$). Regarding Target, accuracy on Unrelated targets was higher than on either Identity targets ($\beta = -3.13$, $SE = 0.30$, $z = -10.60$, $p < .001$) or Competitor targets ($\beta = -4.46$, $SE = 0.31$, $z = -14.16$, $p < .001$). Regarding Exposure, accuracy was higher in the Test-only condition than in the Control-Invariant condition ($\beta = -0.43$, $SE = 0.20$, $z = -2.11$, $p = .035$). In addition, we observed larger differences in accuracy with Indirect-Variant exposure for both the Unrelated-Competitor contrast ($\beta = -2.66$, $SE = 1.25$, $z = -2.13$, $p = .033$) and the Unrelated-Identity contrast ($\beta = -2.45$, $SE = 1.16$, $z = -2.10$, $p = .036$) compared to the Test-only condition. Summary statistics by Target and Exposure are shown in Table 2 and Fig. 3.

For RT, we observed a main effect of Target ($\chi^2(2, N = 3) = 504.91$, $p < .001$). The main effect of Target reflected faster RTs on Unrelated targets than on either Identity targets ($\beta = 15.28$, $SE = 7.05$, $z = 2.17$, $p = .030$) or Competitor targets ($\beta = 180.12$, $SE = 8.19$, $z = 21.99$, $p < .001$). In addition, the Test-only condition was associated with overall faster RTs than the Control-Invariant exposure condition ($\beta = 39.50$, $SE = 15.17$, $z = 2.60$, $p = .009$). Summary statistics by Target and Exposure are shown in Table 2 and Fig. 3.

Table 2
Experiment 1 test task performance: Mean percent correct and mean response time (standard error) by Target for each Exposure group. Accuracy contrasts presented in log odds rather than probability to reflect unit of analysis.

Exposure	*N*	Identity	Competitor	Unrelated	Unrelated vs. Identity	Unrelated vs. Competitor
Accuracy						
Test-only	37	91.96 (1.12)	78.39 (3.82)	99.04 (0.47)	2.196	3.344
Direct-Variant	38	90.99 (1.02)	72.84 (3.96)	99.76 (0.17)	3.713	5.038
Direct-Invariant	42	90.98 (0.89)	71.36 (3.65)	99.06 (0.32)	2.347	3.746
Indirect-Variant	41	90.34 (1.36)	74.11 (3.57)	99.88 (0.12)	4.522	5.705
Indirect-Invariant	41	91.60 (1.09)	74.63 (3.98)	99.24 (0.30)	2.489	3.798
Control-Variant	46	91.71 (1.24)	73.00 (3.83)	99.50 (0.25)	2.882	4.290
Control-Invariant	49	90.09 (1.16)	71.11 (3.72)	98.24 (0.41)	1.816	3.123
Reaction time						
Test-only	37	625 (19)	793 (30)	607 (15)	−17	−186
Direct-Variant	38	646 (24)	820 (31)	600 (14)	−46	−221
Direct-Invariant	42	632 (24)	821 (36)	622 (18)	−9	−199
Indirect-Variant	41	632 (17)	828 (26)	625 (15)	−7	−203
Indirect-Invariant	41	647 (25)	809 (31)	618 (17)	−30	−191
Control-Variant	46	617 (14)	796 (23)	606 (12)	−10	−189
Control-Invariant	49	653 (18)	866 (25)	644 (14)	−9	−222

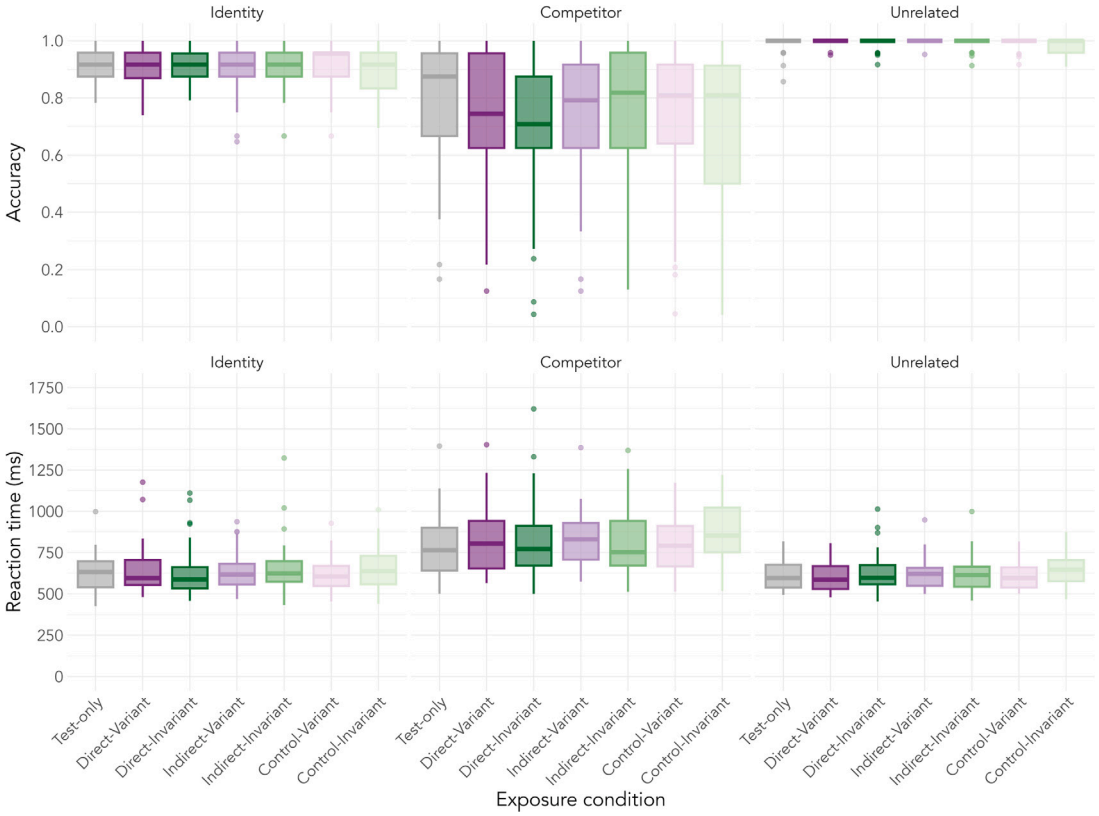


Fig. 3. Experiment 1 test task performance. Top row presents boxplots with mean accuracy by participant. Bottom row presents boxplots with mean reaction times for correct responses by participant.

Test: Comparison between variability and similarity

The accuracy data revealed significant interactions between the Target contrasts and Exposure. Following up this first analysis, we conducted the second analysis focusing on the effects of Variability and Similarity. We observed a main effect of Variability ($\chi^2(1, N = 2) = 9.52$, $p = .002$) that was qualified by an interaction with Target ($\chi^2(2, N = 3) = 9.29$, $p = .010$). The main effect of Variability reflected overall higher accuracy with Variant compared to Invariant exposure ($\beta = 0.57$, $SE = 0.19$, $z = 3.09$, $p = .002$). The Variability-by-Target interaction reflected a larger difference in accuracy for Variant compared to Invariant exposure for both the Unrelated-Identity contrast ($\beta = -1.59$, $SE = 0.52$, $z = -3.04$, $p = .002$) and the Unrelated-Competitor contrast ($\beta = -1.60$, $SE = 0.58$, $z = -2.76$, $p = .006$).

Discussion

Experiment 1 compared the effects of exposure to voiceless stops (Direct), voiced stops (Indirect), or voiceless fricatives (Control) on adaptation to Spanish-accented speech. These three levels of Similarity were crossed with two levels of Variability, such that each talker either produced exemplars of all onsets (Variant) or exemplars of one particular set of onsets (Invariant). Performance on the exposure task reflected a general bias toward labeling tokens as pseudowords rather than as real words; however, listeners were able to distinguish between Spanish-accented real words and pseudowords regardless of exposure condition.

Contrary to our predictions, exposure to Spanish-accented speech did not unequivocally benefit performance on the test task. We observed an increase in the activation of competing targets with Indirect-Variant exposure, with larger differences in accuracy between Competitor and Unrelated targets (*park-bark* vs. *park-wand*) compared to the Test-only condition. We also observed an increase in the activation of matching targets in the Indirect-Variant condition, with larger differences in accuracy between Identity and Unrelated targets (*park-park* vs. *park-wand*) compared to the Test-only condition. Focusing on the unique effects of Variability and Similarity, we found that Variant exposure was associated with increases in both lexical activation and competition compared to Invariant exposure. Overall, our results suggest that listeners responded to Variant exposure by considering more lexical candidates as potential matches to Spanish-accented primes.

This pattern of results contrasts with that of Xie and Myers (2017), which showed that listeners fine-tuned their phonetic categories in response to L2-accented speech. A potential difference in our experimental design was in the construction of the pseudowords. Specifically, we included pseudowords with the experimental onsets that participants were adapting to during exposure. For example, Direct exposure included both the real word *peanut* and the pseudoword **pachine*. While we reasoned that pseudowords like **pachine* should not disrupt perceptual learning of the relevant VOT-/p/ mapping, it is possible that exposure to these mappings in pseudoword contexts disrupted the benefits of exposure to these mappings in disambiguating lexical contexts. To address this issue, we removed these pseudowords in Experiment 2. Instead, participants in the Direct and Indirect groups were both exposed to pseudowords with control onsets like **fachine*. Including control onsets within the Direct and Indirect groups lessened the need for separate Control exposure conditions. To reduce the complexity of the design and home in on the effects of Direct versus Indirect exposure, the Control level of Similarity was not included in Experiment 2.

Experiment 2: Investigating the specificity of generalization from exposure to Spanish-accented stops

Methods

This experiment was the same as Experiment 1, with the exception of the Similarity factor. The Control level was removed, leaving just two levels of Similarity: Direct and Indirect. This change left four between-subjects conditions for the exposure phase: Direct-Variant, Direct-Invariant, Indirect-Variant, and Indirect-Invariant.

Participants

We recruited 192 participants through Prolific, none of whom had participated in Experiment 1. All aspects of recruitment and criteria for data exclusion were the same as in Experiment 1. After removing ineligible participants, 160 remained. After removing participants with poor data quality, 154 remained for analysis (Age: M = 31, SD = 6, Min = 18, Max = 40; Sex: Female = 65, Male = 89; Race: Asian = 5, Black = 17, Multiple selected = 12, Other = 4, White = 116). Data from the participants who completed the experiment without the exposure phase in Experiment 1 were used for comparison.

Materials and procedure

The talkers, tasks, and procedure were the same as those in Experiment 1. Regarding the stimuli, the 288 filler items remained the same. The 72 experimental real words for each level of Similarity (Direct and Indirect) also remained the same. The 72 experimental pseudowords, however, were changed. Recall that in Experiment 1, the experimental pseudowords had the same onsets as the experimental real words. It is possible that this design disrupted adaptation to Spanish-accented VOTs by associating them with pseudowords. To improve learning, we replaced the Direct and Indirect experimental pseudowords with the Control experimental pseudowords from Experiment 1. As a result, in Experiment 2, all experimental pseudowords had control onsets.

Talker assignment and counterbalancing followed Experiment 1. This resulted in 16 experimental lists for the exposure phase, one for each combination of exposure condition (4) and talker assignment (4). All aspects of the test phase remained the same.

Table 3
Experiment 2 exposure task performance: Mean percent correct (standard error) by Word type for each combination of Similarity and Variability. Contrast presented in log odds rather than probability to reflect unit of analysis.

Similarity	Variability	*N*	Real words	Pseudowords	Real words vs. Pseudowords
Direct	Invariant	36	79.36 (1.44)	87.46 (1.86)	−0.595
Direct	Variant	40	79.89 (1.51)	86.87 (1.84)	−0.511
Indirect	Invariant	39	79.50 (1.30)	88.54 (1.35)	−0.690
Indirect	Variant	39	76.78 (1.21)	90.63 (0.98)	−1.074

Analysis and predictions

The data processing, model fitting, and analysis approaches were the same as in Experiment 1; the only change was implementing sum contrasts for the two levels of Similarity. Prior to analyzing exposure task performance for real words with experimental onsets, we removed responses with RTs less than 50 ms (N = 22; 0.10%). We then detected and removed outliers (N = 995; 4.49%). Prior to analyzing test task performance for critical prime-target pairs, we removed responses with RTs less than 50 ms (N = 7; 0.05%). We then detected and removed outliers (N = 679; 4.94%).

Results

Exposure

There was a main effect of Word type ($\chi^2(1, N = 2) = 23.46$, $p < .001$) that reflected higher overall accuracy on pseudowords compared to real words ($\beta = -0.94$, $SE = 0.20$, $z = -4.84$, $p < .001$). Summary statistics for each exposure condition are shown in Table 3.

Test: Comparison to the test-only group

For accuracy, there was a main effect of Target ($\chi^2(2, N = 3) = 190.63$, $p < .001$), with higher accuracy on Unrelated targets than on either Identity targets ($\beta = -2.52$, $SE = 0.27$, $z = -9.42$, $p < .001$) or Competitor targets ($\beta = -3.85$, $SE = 0.28$, $z = -13.81$, $p < .001$). For RT, there was a main effect of Target ($\chi^2(2, N = 3) = 284.08$, $p < .001$), which reflected faster RTs on Unrelated targets than on Competitor targets ($\beta = 161.79$, $SE = 9.76$, $z = 16.57$, $p < .001$). Summary statistics by Target and exposure condition are shown in Table 4 and Fig. 4.

Discussion

In Experiment 2, two levels of Similarity—Direct and Indirect—were crossed with two levels of Variability—Variant and Invariant—in order to investigate how exposure to Spanish-accented speech transfers to a novel talker. We provided listeners with even clearer exposure to Spanish-accented stops compared to Experiment 1 by changing the composition of the pseudowords. We still observed an overall bias toward labeling tokens as pseudowords, but listeners were able to distinguish effectively between the two word types during learning.

Removing the pseudowords with experimental onsets from the exposure task also removed the (undesirable) increases in lexical competition that we observed in Experiment 1. In fact, we did not observe any differences in performance between the Test-only and exposure conditions. This lack of an effect of exposure does not align with the wealth of evidence showing that experience with an L2 accent improves perception of a novel L2-accented talker with the same L1

Table 4
Experiment 2 test task performance: Mean percent correct and mean response time (standard error) by Target for each Exposure group. Accuracy contrasts presented in log odds rather than probability to reflect unit of analysis.

Exposure	*N*	Identity	Competitor	Unrelated	Unrelated vs. Identity	Unrelated vs. Competitor
Accuracy						
Test-only	37	91.96 (1.12)	78.39 (3.82)	99.04 (0.47)	2.196	3.344
Direct-Variant	40	91.71 (1.26)	72.29 (3.69)	98.67 (0.42)	1.900	3.346
Direct-Invariant	36	90.76 (1.27)	80.65 (2.88)	99.01 (0.39)	2.324	3.181
Indirect-Variant	39	91.10 (1.46)	73.41 (3.57)	99.22 (0.27)	2.525	3.835
Indirect-Invariant	39	91.91 (1.15)	72.97 (3.92)	99.51 (0.24)	2.881	4.318
Reaction time						
Test-only	37	625 (19)	793 (30)	607 (15)	−17	−186
Direct-Variant	40	630 (14)	821 (31)	629 (16)	−1	−192
Direct-Invariant	36	687 (26)	798 (29)	641 (17)	−46	−158
Indirect-Variant	39	614 (19)	827 (43)	610 (19)	−4	−217
Indirect-Invariant	39	653 (26)	793 (25)	628 (19)	−26	−166

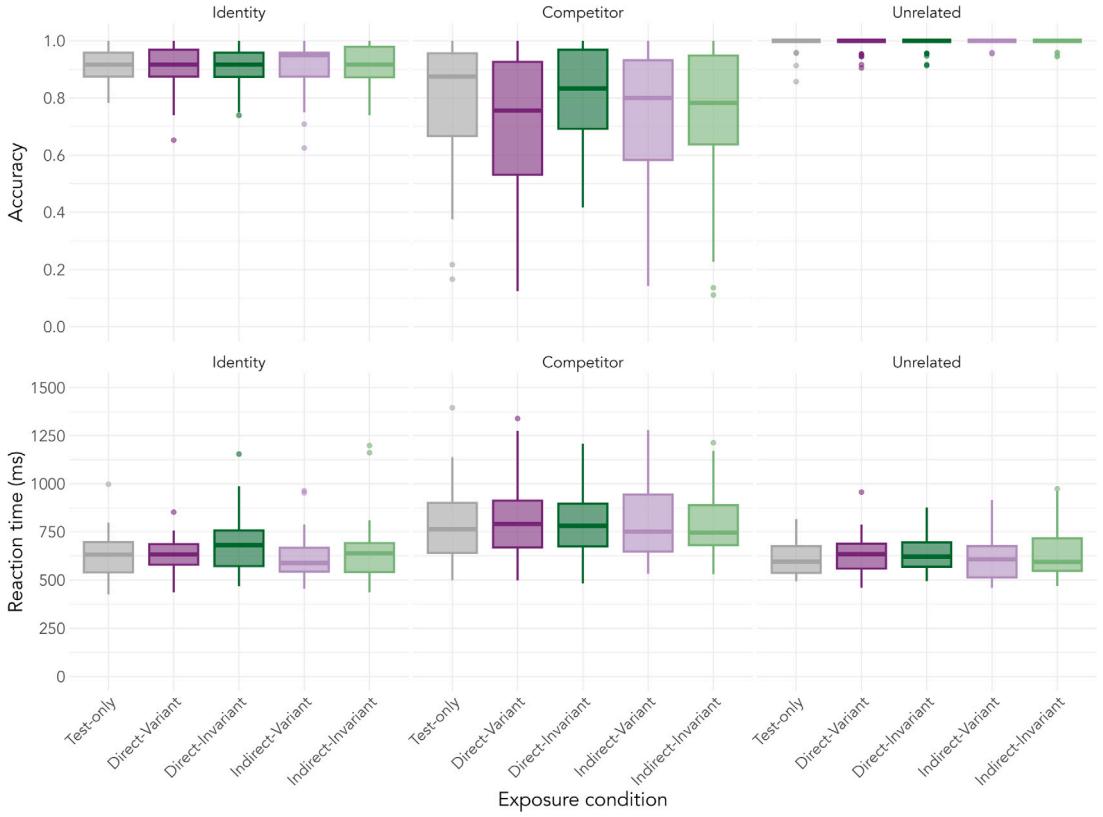


Fig. 4. Experiment 2 test task performance. Top row presents boxplots with mean accuracy by participant. Bottom row presents boxplots with mean reaction times for correct responses by participant.

(Bent & Baese-Berk, 2021). Previous studies have generally used either sentence transcription (e.g., Bradlow & Bent, 2008) or primed lexical decision (e.g., Xie & Myers, 2017) to test the strength of adaptation. In Experiments 1 and 2, we used a matching task under the assumption that categorization of short lag VOTs as voiceless stops should change as a function of perception of short lag VOTs as voiceless stops. For example, consider the auditory prime *park* and the visual target *bark*. Participants needed to decide whether the onset of the token they had heard was a /b/ or not. If they accurately perceived the onset as a /p/, then they would correctly reject *bark* as a match. Thus, accuracy on the matching task was the outcome of the categorization process. It is possible that this outcome-based measure was not fine-grained enough to capture subtle changes in perception. To better assess changes in perception, we changed the test task to a cross-modal primed lexical decision task for Experiment 3.

Experiment 3: Investigating fine-grained perceptual changes from exposure to Spanish-accented stops

Methods

The exposure phase was the same as Experiment 2, but the task used in the test phase (cross-modal primed lexical decision task) was different. To better detect subtle changes in perception as a function of exposure to Spanish-accented speech, we implemented the primed cross-modal lexical decision task from Xie and Myers (2017). We maintained the same three types of Target: Identity, Competitor, and Unrelated. However, participants performed a different task with these targets relative to Experiment 2. Specifically, participants decided whether the visual target was a real English word or not. The auditory primes should increase or decrease RTs on the visual targets as a function of perceptual adaptation to Spanish-accented voiceless stops.

Participants

We recruited 195 participants through Prolific; none had participated in Experiments 1 or 2. All aspects of recruitment and criteria for data exclusion were the same as in Experiments 1 and 2. After removing ineligible participants, 170 remained. After removing participants with poor data quality, 158 remained for analysis (Age: M = 31, SD = 6, Min = 18, Max = 40; Sex: Female = 80, Male = 78; Race: Asian = 4, Black = 20, Multiple selected = 13, Other = 4, White = 117).

Additionally, a new group of 50 participants was also recruited to complete the experiment without the exposure phase. There were 43 participants remaining after checking the eligibility criteria, and 42 participants remained after checking for data quality (Age: M = 30, SD = 5, Min = 21, Max = 40; Sex: Female = 20, Male = 22; Race: Black = 3, Multiple selected = 5, White = 34).

Stimuli

The real words and pseudowords in the exposure task were the same as those in Experiment 2. The auditory primes in the test task were also the same. The only difference was in the visual targets. The unrelated target for each filler prime (144) was replaced with a pseudoword with a different filler onset from the prime. Three additional pseudoword targets were selected for practice. Stimulus selection is described in Appendix C.

Experimental lists

The experimental lists for the exposure task were the same as in Experiment 2. Talker assignment was also counterbalanced the same way.

For the critical primes (72), the combinations of auditory prime and visual target were counterbalanced across participants in three experimental lists as in Experiment 2. For the filler primes (144), perfect counterbalancing across these three lists was not possible, since three quarters of the filler items in each list (108) needed to have unrelated pseudoword targets. Three sets of primes, divided evenly by onset, were rotated through the assignment of Identity or Unrelated (pseudoword) target as evenly as possible. This resulted in 12 experimental lists for the test phase, one for each combination of critical prime-target pair (3) and talker assignment (4).

Tasks and procedure

The headphone check, exposure task, and post-experiment questionnaire were the same as Experiment 2. The procedure was also the same. The only difference was in the structure of the test task.

The test phase featured the cross-modal primed lexical decision task from Xie and Myers (2017). On each trial, participants first heard a real word (auditory prime) and then saw a real word or pseudoword written on the screen (visual target). They indicated whether the visual target was a real English word or not by pressing the *d* or *k* key on their keyboard. The real word response was mapped to the same key as in the exposure task. Participants completed six practice trials followed by 216 main trials presented in random order. Half of the practice trials (3) and half of the main trials (108) required real word responses. All of the trials with critical primes (72) required real word responses.

Analysis

The data processing, model fitting, and analysis approaches were the same as in Experiment 2. Prior to analyzing exposure task performance, we removed responses with RTs less than 50 ms (N = 10; 0.04%). We then detected and removed outliers (N = 1000; 4.40%). Prior to analyzing test task performance for critical prime-target pairs, we removed responses with RTs less than 50 ms (N = 53; 0.37%). We then detected and removed outliers (N = 1076; 7.50%).

Results

Exposure

There was a main effect of Word type ($\chi^2(1, N = 2) = 33.63$, $p < .001$) that was qualified by an interaction with Variability ($\chi^2(1, N = 2) = 4.82$, $p = .028$). Pseudoword accuracy was overall higher than real word accuracy ($\beta = -1.13$, $SE = 0.20$, $z = -5.80$, $p < .001$). The interaction with Variability reflected a smaller difference in accuracy between word types with Invariant compared to Variant exposure ($\beta = -0.36$, $SE = 0.17$, $z = -2.20$, $p = .028$). Summary statistics by Word type and exposure condition are shown in Table 5.

Table 5
Experiment 3 exposure task performance: Mean percent correct (standard error) by Word type for each combination of Similarity and Variability. Contrast presented in log odds rather than probability to reflect unit of analysis.

Similarity	Variability	N	Real words	Pseudowords	Real words vs. Pseudowords
Direct	Invariant	41	80.83 (1.18)	86.79 (1.39)	−0.444
Direct	Variant	43	80.03 (1.34)	89.11 (1.40)	−0.714
Indirect	Invariant	39	79.46 (0.99)	88.98 (1.10)	−0.736
Indirect	Variant	35	77.54 (0.95)	90.64 (1.21)	−1.031

Test: Comparison to the test-only group

For accuracy, there was a main effect of Target ($\chi^2(2, N = 3) = 109.15$, $p < .001$) that was qualified by an interaction with Exposure ($\chi^2(8, N = 9) = 27.25$, $p < .001$). The main effect of Target reflected higher accuracy on Identity targets ($\beta = 3.28$, $SE = 0.32$, $z = 10.28$, $p < .001$) and Competitor targets ($\beta = 1.47$, $SE = 0.27$, $z = 5.44$, $p < .001$) compared to Unrelated targets. The interaction with Exposure was driven by the Indirect-Invariant condition, which was associated with larger differences in accuracy for both the Unrelated-Identity contrast ($\beta = 1.61$, $SE = 0.56$, $z = 2.88$, $p = .004$) and the Unrelated-Competitor contrast ($\beta = 0.84$, $SE = 0.34$, $z = 2.45$, $p = .014$) compared to the Test-only condition. Summary statistics by Target and Exposure are shown in Table 6 and Fig. 5.

For RT, there were main effects of Target ($\chi^2(2, N = 3) = 109.15$, $p < .001$) and Exposure ($\chi^2(4, N = 5) = 37.05$, $p < .001$) that were qualified by an interaction ($\chi^2(8, N = 9) = 16.54$, $p = .035$). Compared to Unrelated targets, responses were overall faster for both Identity targets ($\beta = -132.96$, $SE = 8.78$, $z = -15.14$, $p < .001$) and Competitor targets ($\beta = -28.02$, $SE = 8.60$, $z = -3.26$, $p = .001$). Compared to the Test-only condition, responses were overall slower in the all of the exposure conditions: Direct-Variant ($\beta = 52.84$, $SE = 13.51$, $z = 3.91$, $p < .001$), Direct-Invariant ($\beta = 81.28$, $SE = 13.79$, $z = 5.89$, $p < .001$), Indirect-Variant conditions ($\beta = 54.81$, $SE = 14.27$, $z = 3.84$, $p < .001$), and Indirect-Invariant ($\beta = 42.30$, $SE = 13.89$, $z = 3.04$, $p = .002$). The Exposure-by-Target interaction was driven by the Indirect-Invariant condition, with larger RT differences between Unrelated and Identity targets compared to the Test-only condition ($\beta = -41.32$, $SE = 15.91$, $z = -2.60$, $p = .009$). Summary statistics by Target and Exposure are shown in Table 6 and Fig. 5.

Test: Comparison between variability and similarity

For accuracy, we observed an interaction between Similarity and Target ($\chi^2(2, N = 3) = 13.55$, $p = .001$), which reflected a smaller difference between Unrelated and Competitor targets for Direct compared to Indirect exposure ($\beta = -0.89$, $SE = 0.24$, $z = -3.63$, $p < .001$). In addition, there was a larger difference between Unrelated and Identity targets for Invariant compared to Variant exposure ($\beta = -1.00$, $SE = 0.40$, $z = -2.49$, $p = .013$).

For RT, we observed interactions between Variability and Target ($\chi^2(2, N = 3) = 7.82$, $p = .020$) and between Variability and Similarity ($\chi^2(1, N = 2) = 6.47$, $p = .011$). The Variability-by-Target interaction reflected larger RT differences between Unrelated and Identity targets for Invariant compared to Variant exposure ($\beta = 27.90$, $SE = 11.18$, $z = 2.50$, $p = .013$). The Variability-by-Similarity interaction reflected a larger difference in RTs between Invariant and Variant exposure in the Direct compared to the Indirect conditions ($\beta = -55.23$, $SE = 21.71$, $z = -2.54$, $p = .011$), with the overall fastest RTs in the Direct-Variant

Table 6
Experiment 3 test task performance: Mean percent correct and mean response time (standard error) by Target for each Exposure group. Accuracy contrasts presented in log odds rather than probability to reflect unit of analysis.

Exposure	*N*	Identity	Competitor	Unrelated	Unrelated vs. Identity	Unrelated vs. Competitor
Accuracy						
Test-only	42	96.51 (0.96)	92.40 (1.46)	84.38 (2.69)	−1.633	−0.810
Direct-Variant	43	97.08 (0.66)	90.93 (1.06)	84.74 (2.36)	−1.790	−0.591
Direct-Invariant	41	97.28 (0.69)	89.90 (1.51)	81.67 (2.59)	−2.083	−0.692
Indirect-Variant	35	96.47 (0.71)	92.96 (1.42)	81.42 (3.55)	−1.831	−1.103
Indirect-Invariant	39	98.25 (0.40)	92.65 (1.24)	77.18 (4.03)	−2.812	−1.315
Reaction time						
Test-only	42	583 (12)	674 (14)	731 (26)	148	57
Direct-Variant	43	604 (19)	692 (22)	722 (21)	118	30
Direct-Invariant	41	626 (20)	755 (25)	783 (30)	157	28
Indirect-Variant	35	617 (19)	707 (18)	762 (21)	145	55
Indirect-Invariant	39	589 (14)	708 (17)	749 (21)	160	41

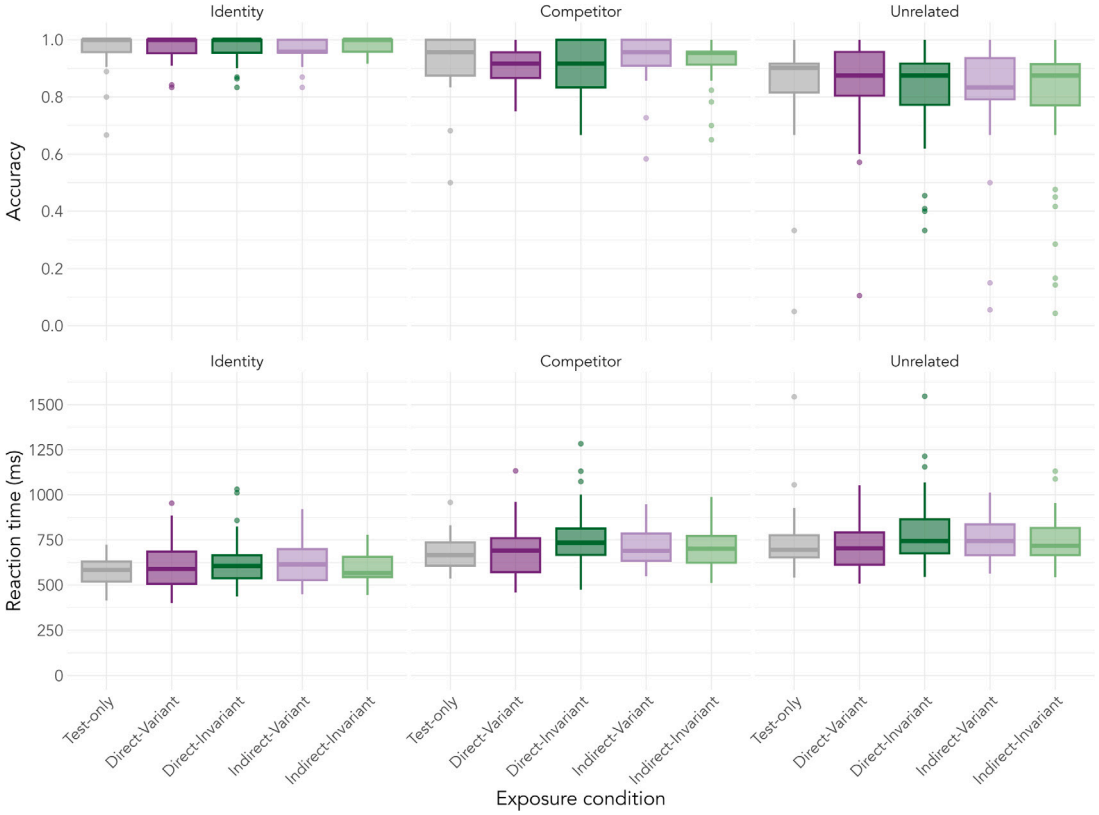


Fig. 5. Experiment 3 test task performance. Top row presents boxplots with mean accuracy by participant. Bottom row presents boxplots with mean reaction times for correct responses by participant.

condition and the overall slowest in the Direct-Invariant condition. In addition, RT differences between Unrelated and Competitor targets were smaller for Direct compared to Indirect exposure (β = 21.81, *SE* = 10.83, z = 2.01, p = .044).

Discussion

Experiment 3 followed up on the findings of Experiment 2 with a different test task. In Experiment 2, we used a matching task during the test phase, which required participants to explicitly compare the visual target to the auditory prime. We did not observe any differences between the Test-only condition and the exposure conditions in this task. In Experiment 3, we took a more implicit approach by using a priming task during the test phase, which probed the extent to which the auditory primes increased or decreased activation of the visual targets. In this way, we could investigate changes in the perception of Spanish-accented English stops.

During exposure, we observed similarly high performance across conditions. Invariant exposure was associated with slightly better performance than Variant exposure. During test, Indirect-Invariant exposure elicited the strongest priming effects compared to the Test-only condition. Comparing the Indirect-Invariant and Test-only conditions, we observed greater identity priming on both the accuracy and RT measures, indicating an increase in lexical activation for voiceless stop onsets. We also observed greater competitor priming with Indirect-Invariant exposure, indicating an increase in lexical competition between voiced and voiceless stops. Together, these results suggest that Indirect-Invariant exposure led to a relaxation of phonetic category criteria for voiceless stops.

When we looked at the exposure conditions alone, we saw that Direct exposure reduced competitor priming relative to Indirect exposure in terms of both accuracy and RT. We also found that Invariant exposure increased identity priming relative to Variant exposure on both measures. While the Direct-Invariant condition itself did not

differ from the Test-only condition, our results suggest that Direct and Invariant exposure are associated with phonetic recalibration. Consider the auditory prime *park* and the visual target *bark*, which are minimal pairs that differ only in the voicing of their onsets. The less the onset of *park* is incorrectly perceived as a /b/, the less it will activate the competitor *bark*. This means that performance on targets like *bark* will become more similar to performance on targets like *wand* that only share a vowel with the prime. We found that Direct exposure induced this shift in the perception of Spanish-accented VOTs, with Spanish-accented *park* being perceived less like *bark*. However, this decrease in lexical competition was not accompanied by an increase in lexical activation. Consider the auditory prime *park* and the visual target *park*; the more the onset of *park* is correctly perceived as a /p/, the more it will activate the matching target *park*. This means that performance on targets like *park* will become less similar to performance on targets like *wand* that only share a vowel with the prime. We found that Invariant exposure was associated with this increase in lexical activation, such that Spanish-accented *park* was perceived more like *park*. Overall, Variability and Similarity had differential impacts on lexical activation and competition.

General discussion

The current study investigated generalization of exposure to L2-accented speech. We compared two hypotheses with different explanations for how exposure supports talker-independent adaptation. The exposure-to-variability hypothesis argues that increasing covariation between acoustic-phonetic cues during exposure increases performance on a new talker (Baese-Berk et al., 2013; Bradlow & Bent, 2008). The similarity-based hypothesis argues that increasing cue-category overlap between the exposure and test talkers increases performance on the test talker (Xie & Myers, 2017). We contextualized these hypotheses in terms of the ideal adapter framework to investigate their explanatory power within the same theoretical model (Kleinschmidt & Jaeger, 2015). Under this theory, listeners develop mental models of the mappings between acoustic cues (like VOT) and phonetic categories (like /p/, /t/, and /k/) according to informative and useful social (here, L2 accent) and indexical (talker) groupings (Kleinschmidt, 2019). To restate the two competing hypotheses within this framework, exposure to variability enhances the development of robust talker-independent generative models, while exposure-test similarity facilitates the selection of the most appropriate generative models (whether talker-specific or talker-independent). Using a novel experimental approach that improved the scope and precision of how variability and similarity are operationalized, we conducted a series of three experiments. The combined results provide some evidence for the similarity-based hypothesis and other evidence against the exposure-to-variability hypothesis.

The key findings come from Experiment 3. After exposure to three Spanish-accented English talkers, we measured performance on a novel Spanish-accented English talker with a primed cross-modal lexical decision task. The critical items in this task were monosyllabic auditory primes with voiceless stop onsets: for example, *park*, *tune*, and *coal*. Importantly, these items had onset competitors with the same place and manner of articulation: *bark*, *dune*, and *goal*, respectively. In Spanish-accented speech, voiceless stops can be perceptually confusable with their voiced stop pairs due to differences in VOT from L1 English norms. Participants made lexical decisions on the visual targets that followed these critical primes. Performance on the different target types indexed different aspects of perceptual adaptation. Specifically, priming of Identity targets indexed lexical activation of the intended word, while priming of Competitor targets indexed lexical activation of perceptually similar words. For example, consider the auditory prime *park*. The Identity target for this prime is *park*, while the Competitor target for this prime is *bark*. If perceptual adaptation to Spanish-accented speech increases activation of the lexical item *park* upon hearing *park*, then responses to the visual target *park* would increase in accuracy and speed. If perceptual adaptation to Spanish-accented speech decreases activation of the lexical item *bark* upon hearing *park*, then responses to the visual target *bark* would decrease in accuracy and speed. Ideally, exposure would both increase activation of the intended word and decrease activation of competing words. We found that Indirect-Invariant exposure increased activation of both voiceless (Identity) and voiced (Competitor) targets compared to no exposure. Comparing Variability and Similarity directly, we found that Direct exposure was associated with a decrease in lexical competition, while Invariant exposure was associated with an increase in lexical activation. To better understand these effects, we relate our findings to the exposure-to-variability and similarity-based hypotheses.

Exposure-test similarity

The similarity-based hypothesis comes from Xie and Myers (2017). In this study, two groups of participants trained on multiple Mandarin-accented talkers. The experimental group was exposed to disambiguating lexical contexts for word-final /d/ (e.g., *overload*), while the other was not. During test on a novel Mandarin-accented talker, lexical activation for /d/-final primes was higher in the experimental group. Among the Mandarin-accented talkers in the training set, one talker's productions of /d/ were similar to those of the test talker on multiple acoustic measures. There was also one talker out of the five whose productions of /d/ differed from the test talker's on these metrics. After single-talker exposure to the similar talker, lexical activation for /t/-final competitors was lower in the experimental group. Multi-talker exposure including the similar talker also facilitated generalization to a novel talker. By contrast, single-talker exposure to the dissimilar talker did not reduce lexical competition. Overall, the specific way that an L2-accented talker produces an L2 variety determines the level of generalization to other talkers. This hypothesis is supported by work on lexically-guided perceptual retuning showing different patterns of generalization for different phonetic contrasts (Kraljic & Samuel, 2006, 2007; Reinisch & Holt, 2014).

To operationalize similarity in our study, we exposed participants to different sets of Spanish-accented stops through an auditory lexical decision task. In Direct conditions, participants gained experience with /p/, /t/, and /k/ onsets in the context of real words like *peanut*, *terminal*, and *kingdom*, respectively. Exposure to Spanish-accented voiceless stops trained listeners to shift the VOT distributions for these sound categories from long lag (English-like) toward short lag (Spanish-like), as illustrated in Fig. 1. In Indirect conditions, participants gained experience with /b/, /d/, and /g/ in the context of real words like *beehive*, *desert*, and *gallop*, respectively. Exposure to Spanish-accented voiced stops trained listeners to shift the VOT distributions for these sound categories from short lag (English-like) toward lead (Spanish-like). Shifting the VOT distribution for voiced stops leftward on the cue continuum had the potential to trigger a more general leftward shift that would move the VOT distributions for voiceless stops from long lag to short lag.

Our approach to investigating similarity involved exposure to different sound categories. By contrast, the approach of Xie and Myers (2017) involved exposure to different cue distributions over the same sound category. That study compared the means of three different cues to voicing in word-final position to determine which talker had the most (dis)similar cue-category distributions to the test talker; separate groups of participants were then exposed to either the similar talker or the dissimilar talker. As a result, their design featured different talkers across exposure conditions. Our design featured the same talkers across exposure conditions, allowing us to control the amount of talker-specific acoustic-phonetic variability between levels of Similarity. In turn, this allowed us to manipulate the constructs of Similarity and Variability independently. We did not, however, control the exact VOT-stop distributions that listeners heard during exposure. Rather, we took advantage of the fact that the VOT distributions for the /p/, /t/, and

/k/ onsets in the Direct exposure conditions were not only distinct from those for the /b/, /d/, and /g/ onsets in the Indirect exposure conditions, but also similar to those for the critical test items.

Experiment 3 provided some support for the similarity-based hypothesis. The similarity-based hypothesis predicted a main effect of Similarity, with better test performance after Direct compared to Indirect exposure. Participants in the Direct condition were also expected to outperform the Test-only group. Specifically, Direct exposure was expected to increase identity priming and decrease competitor priming. We found that Direct exposure decreased lexical competition compared to Indirect exposure consistent with phonetic recalibration from exposure to /p/, /t/, and /k/ onsets. However, neither of the Direct exposure groups (Direct-Variant or Direct-Invariant) differed from the Test-only group. By comparison, Indirect-Invariant exposure was associated with increased competitor priming compared to the Test-only group, as well as increased identity priming. Instead of facilitating generalization through fine-tuning phonetic categories for Spanish-accented speech, the Indirect-Invariant condition simply increased the number of lexical candidates under consideration. These results suggest that exposure-test similarity alone is not sufficient for generalization; variability during exposure is also a key consideration.

Exposure to variability

The exposure-to-variability hypothesis is based on the findings of Baese-Berk et al. (2013). In that study, exposure to multiple talkers, each with a different L2 accent, generalized to a novel talker with an unfamiliar L2 accent. Exposure to multiple talkers with the same L2 accent did not generalize to an unfamiliar L2 accent (Bradlow & Bent, 2008). However, both types of exposure generalized to a novel talker with a familiar L2 accent. The multi-accent condition included one Mandarin-accented talker out of five talkers with different L1s, while the single-accent condition included five Mandarin-accented talkers. In both cases, performance on a novel Mandarin-accented talker was higher than in the control condition, suggesting that listeners benefit from exposure to systematic covariation across L2 accents. Overall, the way that L2 and L1 varieties differ at a high level generalizes across L2 talkers. This hypothesis is also supported by recent work on cross-accent generalization (Bradlow et al., 2023), as well as by the large body of work on high variability phonetic training for L2 acquisition (for a review, see Zhang et al., 2021).

To operationalize variability in our study, we exposed participants to different combinations of talker and onset. In Variant conditions, all three exposure talkers produced all three onsets in an experimental group. For example, consider the critical onset /p/ in items like *peanut*, *palace*, and *pianist*. Participants in the Direct-Variant exposure group might hear Talker A produce *peanut*, Talker B produce *palace*, and Talker C produce *pianist*. As a result, participants would gain experience with each talker's VOT distribution for /p/. This condition was designed to encourage the formation of talker-independent generative models for each onset. By contrast, in Invariant conditions, each of the three exposure talkers was assigned to one of the three onsets in an experimental group. This means that participants in the Direct-Invariant exposure group might only hear Talker A produce items with /p/ onsets like *peanut*, *palace*, and *pianist*. This condition was designed to encourage talker-specific generative models for each onset.

Our approach to operationalizing variability allowed us to hold the amount of exposure to each talker, as well as the talkers themselves, constant between conditions. By contrast, Bradlow and Bent (2008)'s approach was to vary the number of talkers encountered during exposure. They found that multi-talker exposure—comprising five L2-accented talkers with a common L1—facilitated talker-independent adaptation more than single-talker exposure—comprising one of the five talkers from the multi-talker condition. The hypothesis was that exposure to multiple talkers would help listeners distinguish the common features of an L2 accent from the specific features of an L2 talker by increasing variation in the input. However, subsequent studies have shown that the benefit of multi- over single-talker exposure depends on the specific combination of exposure and test talkers (e.g., Bradlow et al., 2023). To neutralize any talker-specific effects, we counterbalanced the assignment of talkers to the exposure and test phases as in Xie et al. (2021) and Bradlow et al. (2023).

None of the three experiments provided support for the exposure-to-variability hypothesis. The exposure-to-variability hypothesis predicted a main effect of Variability, with better test performance after Variant compared to Invariant exposure. Participants in both conditions were also expected to outperform the Test-only group. In particular, Variant exposure was expected to increase the activation of intended targets and decrease the activation of unintended targets. We found that Variant exposure was associated with increased lexical competition in both Experiments 1 and 3. This suggests that auditory primes like *park* were perceived more like *bark* after exposure to variability. In Experiment 1, Variant exposure was also associated with increased lexical activation, which suggests that the presence of pseudowords with stop consonant onsets during exposure increased uncertainty about these phonetic categories during test. In fact, when these items were removed, the effect of Variability on Identity targets (a) disappeared in Experiment 2 and (b) was reversed in Experiment 3, such that Invariant exposure increased identity priming. Overall, Variant exposure was associated with a relaxation of phonetic category criteria, with listeners considering both /p/ and /b/ onsets for words like *park* (Zheng & Samuel, 2020). The positive association between Invariant exposure and identity priming in Experiment 3 suggests that talker-specific generative models transferred to a novel talker more readily than talker-independent models. Although we included a random intercept for talker in all of our analyses, it is possible that the amount of variability in stop production between our four talkers made talker-independent models less useful for adaptation (see Table B.8 and Figure B.6 in Appendix B). In future work, manipulating between-talker variability by changing the means and shapes of the VOT distributions will provide further insight into the impact of variability on generalization. Including multiple novel talkers with different acoustic properties in the test phase would be another way of investigating the relations between variability and exposure-test similarity in accent adaptation.

Conclusion

In summary, our results show that talker-independent adaptation to L2-accented speech is facilitated by exposure-test similarity and reducing between-talker variability during exposure. We found that encouraging the development of talker-specific generative models for voiceless stops during exposure reduced competition from voiced stops and increased activation of voiceless stops during test. Our findings are in line with previous work showing that listeners can generalize from exposure to a single talker (Xie et al., 2021; Xie & Myers, 2017). This study is the first to compare variability and similarity while maintaining the same talkers across conditions. This level of control is key to understanding how between-talker variability and talker-specific phonetic detail are used for adaptation. Here, we showed that constraining between-talker variability during exposure allows listeners to use talker-specific phonetic detail during test.

CRediT authorship contribution statement

Holly A. Zaharchuk: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Janet G. van Hell:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Science Foundation, United States (BCS-2234907 to Zaharchuk and Van Hell; BCS-2041264 to Van Hell) and the Pennsylvania State University, United States (University Graduate Fellowship to Zaharchuk). The findings and conclusions do not necessarily reflect the views of the funding agencies.

Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jml.2026.104770.

Data availability

The data and analyses included in this manuscript can be found on the Open Science Framework (https://osf.io/z3q6v). The full list of experimental stimuli and the post-experiment questionnaire can also be found in the repository.

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